

PROJECT REPORT

AI-Powered Sustainable Packaging Recommendation System

AI Framework for Sustainable Packaging Design and Material
Optimization

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Infosys Springboard Virtual Internship (AI Domain)

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Submitted by:

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Mentor/Guide:

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Certificate

This is to certify that **Om Kapse** has successfully completed the project titled “**AI-Powered Sustainable Packaging Recommendation System**” as part of the **Infosys Springboard Virtual Internship (AI Domain)** during January 2026. The project work has been carried out under my guidance and supervision.

Guide Name: Mr. Shubham Lad

Date: 20/01/2025

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I am thankful to my mentor for their continuous support. This project helped me gain practical knowledge in **Machine Learning, Data Processing, Database Management, API Development, Dashboard Analytics, and Cloud Deployment**.

Name: Om Kapse

Date: January 2026

Abstract

Packaging selection plays an important role in logistics and sustainability. The wrong packaging material increases product damage risk, cost, and CO₂ emissions.

This project, **EcoPackAI**, is an AI-powered sustainable packaging recommendation system designed to suggest the best eco-friendly packaging material based on product parameters such as **weight, volume, and fragility**. The system performs preprocessing, feature engineering, and model training using **Random Forest and XGBoost** models. The trained models achieve excellent evaluation results including **R² up to 0.99998**.

The system is developed as a full-stack application with a **Flask backend and web frontend**. Recommendations are served via REST API endpoint **/predict** and a **Business Intelligence Dashboard** is added to visualize CO₂ reduction percentage, cost savings and material usage trends. The complete system is deployed on **Render cloud platform** and connected to **MySQL database**.

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1. Introduction

Packaging is essential for product safety, storage and transportation. Environmental sustainability has become important due to increasing packaging waste and carbon footprint. Industries need an intelligent packaging selection system that balances:

- Cost
- Durability
- Recyclability
- Biodegradability
- CO₂ Emissions

EcoPackAI solves this using Machine Learning and AI-based ranking.

2. Problem Statement

Choosing packaging manually is inefficient and leads to:

- Higher CO₂ emissions
- Increased cost
- Unsuitable material selection for fragile/heavy products

Problem: Develop a system that recommends sustainable packaging materials based on product characteristics and environmental impact.

3. Objectives

3.1 Main Objectives

- Collect packaging material dataset
- Store dataset in MySQL
- Clean dataset and remove outliers
- Perform feature engineering
- Train ML models for predictions
- Create ranking algorithm for recommendation
- Develop Flask API backend
- Develop frontend UI
- Create BI dashboard analytics
- Deploy project on Render cloud with MySQL

4. Scope of the Project

4.1 In Scope

- Sustainable packaging material recommendation
- Cost and CO₂ prediction
- Full-stack integration
- BI dashboard analytics
- Cloud deployment

4.2 Out of Scope (Future)

- Real supplier pricing API integration
- Mobile app + barcode scanning
- Real-time large industrial dataset integration

5. System Requirements

5.1 Hardware Requirements

- Minimum 4GB RAM Laptop/PC
- Stable Internet connection

5.2 Software Requirements

- Python 3.10+
- Jupyter Notebook / VS Code
- MySQL Server
- Git & GitHub
- Render Cloud Platform

6. Methodology

1. Dataset collection
2. Store dataset in MySQL
3. Data preprocessing and cleaning
4. Outlier detection using IQR method
5. Outlier treatment using clipping
6. Feature engineering and encoding
7. Train ML models (Random Forest, XGBoost)
8. Evaluate using RMSE, MAE and R^2
9. Create ranking algorithm
10. Develop Flask API backend
11. Build frontend UI
12. Build BI dashboard analytics
13. Deploy on Render with MySQL cloud DB

7. Module-wise Implementation

7.1 Module 1: Data Collection and Management

- Imported dataset into Pandas
- Connected MySQL using mysql.connector
- Stored dataset into MySQL table

Result: Dataset stored successfully.

7.2 Module 2: Data Cleaning and Feature Engineering

- Checked missing values and duplicates
- Outlier detection using IQR method
- Outlier treatment using clipping
- Exported cleaned dataset

Result: Clean dataset created for ML training.

7.3 Module 3: Dataset Preparation

- Feature selection and target separation
- Train-test split
- Scaling applied using StandardScaler

7.4 Module 4: ML Prediction Models and Ranking System

- Random Forest Regressor used for Cost Prediction
- XGBoost Regressor used for CO₂ Prediction
- AI ranking created using eco score + cost index

7.5 Module 5: Flask Backend API

- Flask backend created with CORS enabled
- REST API endpoint `/predict` implemented
- JSON request/response supported

7.6 Module 6: Frontend UI

- HTML, CSS, JS UI created
- User inputs: weight, volume, fragility
- Results show best recommendation + ranking table

7.7 Module 7: Business Intelligence Dashboard

- Dashboard created to show CO₂ reduction percentage
- Cost savings analytics visualization
- Material usage trends analysis
- Reports exported in Excel (dashboard_report.xlsx, sustainability_report.xlsx)

Result: BI dashboard prepared for sustainability insights.

7.8 Module 8: Deployment and Documentation

- Project deployed on Render
- MySQL connected in deployed environment using environment variables

- API verified after deployment
- Documentation and README completed

Note: Project report and video demo will be added later.

8. Machine Learning Models and Results

8.1 Evaluation Metrics

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- R^2 Score

8.2 Actual Model Results

Random Forest (Cost Prediction)

- RMSE: 0.03425175
- MAE: 0.01396171
- R^2 : 0.98610537

XGBoost (CO₂ Prediction)

- RMSE: 0.00118505
- MAE: 0.00099652
- R^2 : 0.99998250

Conclusion: High accuracy achieved in both models.

9. Backend API Implementation

9.1 Endpoints

Method	Endpoint	Purpose
GET	/	Backend health/status
POST	/predict	Returns best recommendation + ranking

9.2 Example Request

```
POST /predict
{
  "weight": 2.5,
  "volume": 800,
  "fragility": 4
}
```

10. Frontend UI Implementation

Frontend developed using HTML, CSS and JavaScript:

- Input Form for product parameters
- Best Recommendation Display
- Ranking Table Display

11. Deployment

The project was deployed on **Render**:

- Flask backend deployed as Render Web Service
- MySQL cloud database connected
- End-to-end system verified online

12. Conclusion

EcoPackAI successfully demonstrates:

- Dataset preprocessing and feature engineering
- High accuracy ML models
- AI ranking-based recommendation generation
- Flask backend + frontend UI integration
- BI dashboard analytics
- Cloud deployment on Render with MySQL

13. Future Enhancements

- Add more API endpoints: cost and CO₂ separately
- Store recommendation history in database with user accounts
- Improve BI dashboard with filters and PDF reports
- Mobile app integration

14. References

- Infosys Springboard Training Materials
- <https://scikit-learn.org/>
- <https://xgboost.readthedocs.io/>
- <https://flask.palletsprojects.com/>
- <https://dev.mysql.com/doc/>
- <https://render.com/docs>